

Program : Diploma in Automation and Robotics	
Course Code : 6331C	Course Title: Mechatronics and Automation
Semester: 6	Credits: 4
Course Category: Program core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- Understand the mechatronic system, usage and advantages of mechatronics
- Understand the principles of transducers and sensors
- Comprehend the working of Control valves and actuators
- Understand the concepts of Programmable Logic Controllers

Course Outcomes:

On completion of the course, the student will be able to:

Con	Description	Duration (Hours)	Cognitive Level
CO1	Understand the working principle of various sensors and actuators in mechatronics systems	15	Understanding
CO2	Formulate and simulate models of mechatronics systems	15	Applying
CO3	Explain the implementation of PLC in mechatronics applications	13	Applying
CO4	Design and analysis of commonly encountered mechatronics systems for real time applications	15	Applying
	Series Test	2	

CO – PO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3						
CO2	3	2					
CO3	3						
CO4	3						

3-Strongly mapped

2-Moderately mapped

1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the working principle of various sensors and actuators in mechatronics systems.		
M1.01	Illustrate the working of a strain gauged load cell	3	Understanding.
M1.02	Explain the principle of operation and suggest two applications of Hall effect sensor in mechatronic systems.	3	Understanding.
M1.03	With neat sketches explain the working of a double acting hydraulic actuator.	2	Applying
M1.04	Explain the working of any one non-contact temperature measurement system	4	Applying
M1.05	Explain any two situations when pneumatic actuators are preferred over hydraulic ones.	3	Applying
Contents: Introduction to Mechatronics: Structure of Mechatronics system. Comparison between traditional and mechatronics approach. Sensors - Characteristics -Temperature, flow, pressure sensors. Displacement, position and proximity sensing by magnetic, optical, ultrasonic, inductive, capacitive and eddy current methods.			
CO2	Formulate and simulate models of mechatronics systems		
M2.01	Derive the mathematical model of a general electrical system and draw its analogy with a mechanical system.	5	Applying
M2.02	Explain the working of a mechanical device using closed loop control system with the help of a suitable example.	5	Applying
M2.03	Explain the working of electrical drives: DC, AC, brushless, servo and stepper motors	5	
	Series Test – I	1	
Contents: Actuators: Hydraulic and Pneumatic actuators - Directional control valves, pressure control valves, process control valves. Rotary actuators. Development of simple hydraulic and pneumatic circuits using standard Symbols. Electrical drives: DC, AC, brushless, servo and stepper motors.			

CO3	Explain the implementation of PLC in mechatronics applications		
M3.01	Understand the concepts of Programmable Logic Controllers	3	Applying
M3.02	Explain the basic components of PLC with block diagram	3	Applying
M3.03	Explain timers – internal relays and counters	2	Applying
M3.04	Explain selection of PLC	3	Applying
M3.05	List the programming language of PLC	2	Applying
Contents: Programmable Logic Controller definition -block diagram- Input / Output processing – Programming – mnemonics – Timers, internal relays and Counters – shift register – Master and jump controls – Ladder logic circuit -Data handling- selection of PLC – Microprocessor and Microcontrollers – Applications.			
CO4	Design and analysis of commonly encountered mechatronics systems for real time applications		
M4.01	Understand the techniques of Fault finding in mechatronic system	3	Understanding
M4.02	Describe the Fault detection techniques in microprocessor	3	Understanding
M4.03	Explain the possible mechatronic design solutions	5	Applying
M4.04	comprehend the mechatronic system approach with case study	4	Applying
	Series Test – II	1	
Contents: Mechatronics systems and Fault finding Fault finding – Fault detection techniques – common hardware faults. – Possible design solutions -Timed switch – bathroom scales. Course Content Describe Stepper Motor. Mechatronics in Robotics- choice of Sensors and Actuators. Robotic vision system – Image. Acquisition: Vidicon, charge coupled device (CCD) and charge injection device (CID) cameras.			

Text / Reference

T/R	Book Title/Author
T1	Bolton W., Mechatronics: Electronic Control Systems in Mechanical and Electrical Engineering, Person Education Limited, New Delhi, 2007
R1	David G. Aldatore, Michael B. Histan, Introduction to Mechatronics and Measurement Systems, McGraw-Hill Inc., USA, 2003.
R2	Ramachandran K. P., G. K. Vijayaraghavan, M. S. Balasundaram, Mechatronics: Integrated Mechanical Electronic Systems, Wiley India Pvt. Ltd., New Delhi, 2008.
R3	Devdas Shetty, Richard A. Kolk, “Mechatronics System Design”, Thomson Learning Publishing Company, Vikas publishing house, Second edition, 2001.